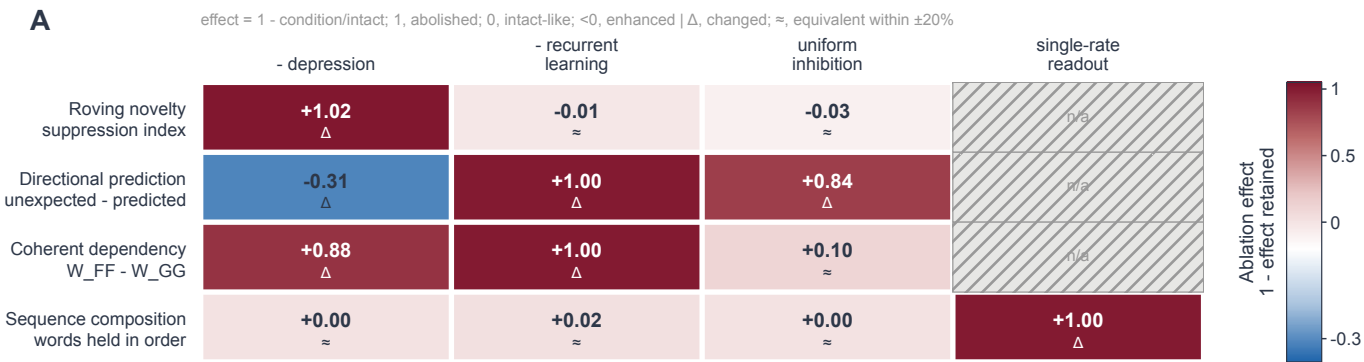


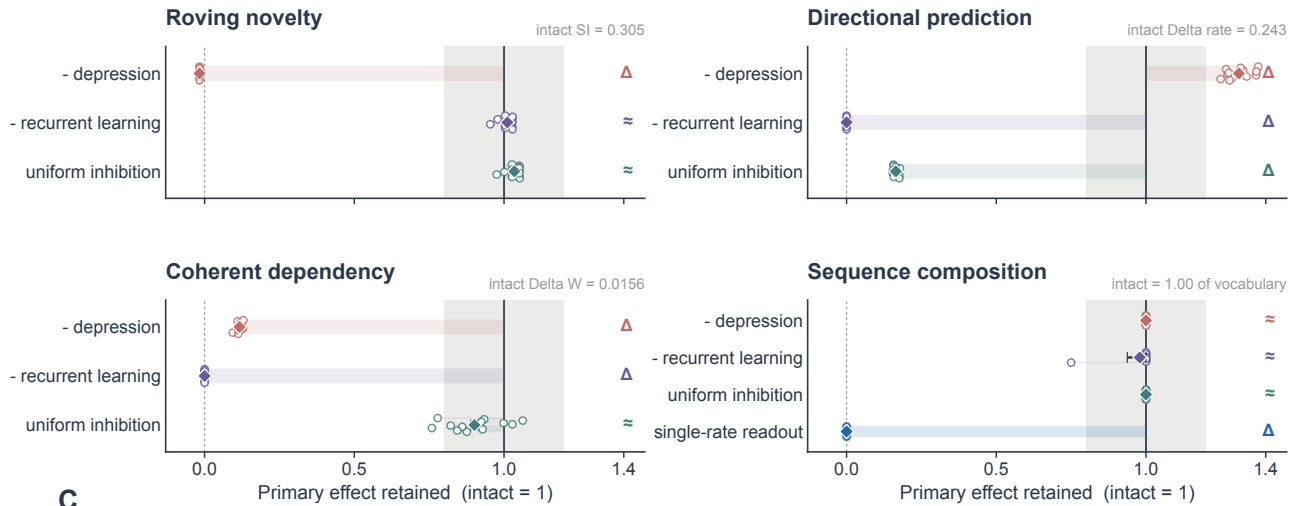
One frozen core reveals a sparse causal sensitivity map

effect = 1 - condition/intact; 1, abolished; 0, intact-like; <0, enhanced | Δ , changed; $\approx$, equivalent within $\pm 20\%$



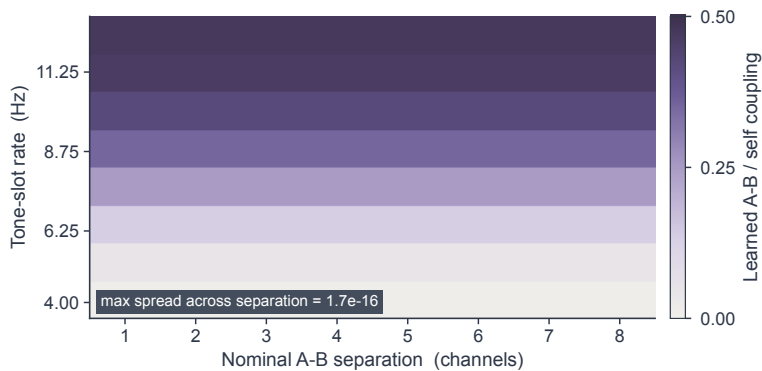
Paired sessions separate statistical change from practical equivalence

n = 12 per task; points are paired sessions, diamonds are means, bars are paired-bootstrap 95% CIs; gray band is the prespecified $\pm 20\%$ SESOI



The no-retuning ABA-silence test has rate sensitivity but no Δf axis

80 ABA-silence cycles per cell; one-hot tones, frozen core, zero initial W; the map reports model coupling, not a perceptual boundary



Permutation audit

$w_{ij} = w_{lat}$ for every $i \neq j$
same w_{lat}



A classic van Noorden boundary cannot emerge until the fixed front end or connectivity contains spectral distance.

D Mechanistic synthesis



A coordinated toolkit is supported within one frozen core; spectral-distance structure remains the explicit missing computation.